Movie Recommendation System

System Architecture Specification & Technical Documentation — Category: Machine Learning & Information Retrieval

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Repository: Pending Repository Verification

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1. Overview & Recommendation Approaches

Recommender systems overcome digital content choice paralysis. This project specifies a content-based movie recommendation engine utilizing natural language processing, TF-IDF vectorization, and cosine similarity across movie metadata attributes (genres, directors, cast members, keywords, and plot summaries).

2. Proposed Algorithmic Architecture

• **Metadata Fusion:** Concatenation of cleaned genres, overview tokens, director names, and top billing cast members into a unified content tag string.

• **Vectorization:** `CountVectorizer` or `TfidfVectorizer` transforming textual tags into high-dimensional sparse token matrices.

• **Similarity Metric:** Pairwise cosine similarity matrix computing angular distances between films.

• **Recommendation Inference:** User selects a movie; system sorts similarity row and returns top-10 nearest neighbor titles.

3. Verification Status

• **Status:** Not documented in current active repository list. Maintained as pending.

Architecture Note: Repository link is kept as an editable pending placeholder per instructions. Implementation details will be linked upon repository upload.